Decoding Light: Physics-Aware Spatial-Spectral Transformer for Low-Cost High-Fidelity Hyperspectral Imaging

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Abstract

Hyperspectral imaging enables simultaneous acquisition of spatial and spectral information and has become indispensable in applications including precision agriculture, material characterization, biomedical imaging, etc. However, widespread adoption remains limited by the lack of cost-effective imaging hardware and accurate reconstruction methods that can reliably recover high-dimensional spectral data. To address this gap, I develop a low-cost Computed Tomography Imaging Spectrometer (CTIS) hardware system that captures the zero, first, and diagonal second order diffraction patterns in a single exposure, together with a novel physics-aware self-supervised transformer framework for high-fidelity hyperspectral reconstruction. Since the CTIS hardware simplification creates an under-determined inverse reconstruction problem with far fewer measurements than unknowns, I formulate its solution as the combination of a physics-based backprojection process and a deep learning process. To ensure physically plausible learning, I design a Physics-Aware Spatial-Spectral (PASS) Transformer that embeds the physics of diffraction and point-spread function operations directly into its attention mechanisms to estimate missing spatial–spectral information using physics-guided learned priors. PASS-Transformer also reconstructs overlapping spectral components inherent in CTIS by enforcing physical consistency, hence reducing hallucinated structures and spatial-spectral ambiguities. Furthermore, by exploiting forward consistency with CTIS imaging physics during training, its self-supervised training approach eliminates the need for costly ground-truth hyperspectral data. Experimental results demonstrate that my method outperforms state-of-the-art techniques in reconstruction fidelity. I further apply and validate this integrated hardware-AI system in plant condition monitoring, environmental contamination analysis, and food allergy detection.

1 Introduction

Hyperspectral imaging (HSI) enables simultaneous acquisition of spatial and spectral information across many narrow wavelength bands and provides detailed spectral signatures that reveal material composition, chemical characteristics, and biological conditions beyond the capabilities of conventional Red-Green-Blue (RGB) imaging. It has become indispensable in applications including robotic vision, environmental engineering, precision agriculture, biomedical imaging, and so on [Bhargava et al., 2024]. Despite its strong potential, HSI remains expensive and relatively inaccessible in practice. Conventional hyperspectral acquisition systems typically rely on scanning across spatial or spectral dimensions [Luo et al., 2021], which requires complex and expensive optical designs, long acquisition times, and accurate mechanical control. These systems are further limited by low temporal resolution and are highly vulnerable to motion artifacts, making them less suitable for dynamic scenes or real-time operation. Snapshot hyperspectral imaging aims to overcome these limitations by encoding spectral information into a single detector exposure [Xiong et al., 2017]. However, snapshot systems also suffer from limitations, such as expensive hardware, compressed and overlapping measurements, and ill-posed reconstruction with incomplete information. Consequently, although snapshot HSI improves imaging speed and mitigates motion artifacts, its broader accessibility is still constrained by its hardware design as well as hyperspectral image reconstruction difficulty.

Among snapshot approaches, computed tomography imaging spectrometer (CTIS) is a simple yet effective hardware solution [Habel et al., 2012, Salazar-Vazquez and Mendez-Vazquez, 2020, Zhou et al., 2024, Wang et al., 2025], which uniquely employs diffraction optics to project multiple spectrally dispersed views of the scene onto a 2D sensor and allows reconstruction of the full hyperspectral data cube via tomographic inversion. However, physical constraints on detector size, signal-to-noise ratio, and optical complexity often limit the number of available diffraction orders, resulting in highly incomplete tomographic measurements. Furthermore, wavelength-dependent point spread functions (PSFs) and two-dimensional (2D) spectral dispersion introduce substantial spatial–spectral coupling, resulting in reconstruction ambiguity. Alternatively, mask-based hyperspectral imaging uses carefully engineered coding schemes to provide compact and flexible sensing designs [Yue et al., 2024, Di et al., 2025], in which coded apertures or compressive masks modulate incoming light to multiplex spatial and spectral information into a single measurement [Shi et al., 2024]. Due to the physical nature of the coded mask, they face practical limitations including reduced light efficiency, amplified noise sensitivity, and reconstruction instability caused by heavily compressed measurements. By comparison, CTIS still offers a more physically transparent measurement model and a direct tomographic interpretation. It is an ideal hardware candidate for physics-based reconstruction.

To reconstruct hyperspectral images from 2D snapshot measurements, traditional model-based reconstruction methods, such as expectation–maximization (EM) [He et al., 2017], maximum likelihood estimation [Johnson et al., 2006], and sparse regularization inversion [Xie et al., 2022], rely on iterative optimization techniques and explicitly invert a physics-based forward model using handcrafted priors. While these methods enforce physical consistency, they are fundamentally limited by the under-determined nature of the inverse problem, resulting in spectral cross-talk and spatial ambiguity under view-limited conditions. Furthermore, they often converge slowly, are sensitive to noise and model mismatch, and struggle to recover fine spatial-spectral details. Recent advances in deep learning have demonstrated promising performance

in hyperspectral reconstruction and inverse problems [Ongie et al., 2020] through generating missing spectral information using learned data distributions or directly mapping sensor measurements to hyperspectral cubes via supervised learning [Miao et al., 2019]. Purely data-driven learning models often ignore the underlying physics of diffraction and image formation, generating hallucinations among spatial and spectral interactions [Zhang et al., 2022].

Therefore, recovering a full hyperspectral data cube from a single exposure measurement in a snapshot HSI system remains a challenging ill-posed inverse problem. In the system, each sensor pixel encodes a multiplexed superposition of spatial and spectral information, resulting from wavelength-dependent diffraction, point-spread function blur, and overlap across diffractions. This many-to-one mapping causes irreversible information loss. Solving this problem requires not only accurate modeling of the image formation but also strong spatial–spectral priors capable of resolving ambiguities inherent in the physical process.

In my point of view, a viable solution requires an HSI hardware and software co-design. Its image reconstruction requires the joint enforcement of physical laws and constraints, including spatial point-spread consistency, spectral diffraction consistency, and forward/backward cyclic consistency, into a data-driven deep learning process. Spatial consistency ensures the mixture of neighboring pixels follows the point spread function (PSF) defined by the snapshot imaging system, while spectral consistency further guarantees that the recovered spectral signatures obey physically valid wavelength-dependent dispersion. Forward/backward cyclic consistency ensures that the reconstructed hyperspectral volume, when forward-projected through the known diffraction imaging model, reproduces the measured multiplexed sensor image. An imaging system and a reconstruction method that jointly satisfy these consistencies can yield physically interpretable and verifiable hyperspectral cubes, which is critical for scientific reliability and high-fidelity analysis.

Building on these insights, in this work I design a novel Physics-Aware Spatial-Spectral Transformer (PASS-Transformer) architecture for an optimized view-limited CTIS hyperspectral imaging system. Rather than learning a direct inverse mapping, the proposed framework uniquely decomposes the reconstruction into a physics-based backprojection followed by learned spatial–spectral refinement. It explicitly embeds the hyperspectral imaging system's point-spread function operation, diffraction process, and optical forward model into the transformer's spatial / spectral attentions and self-supervision learning mechanism. Essentially, the transformer network solves the reconstruction with known physical principles so that it can conquer the inverse problem with faster speed, less training data, and higher accuracy. Starting from a physics-based backprojection initialization, PASS-Transformer performs iterative reconstruction through two stages. In the first stage, the transformer refines the latent hyperspectral representation by learning spatial-spectral correlations upon physics constraints from PSF and diffraction, i.e., following point-spread function and diffraction geometry guided wavelength-dependent dispersion. In the second stage, PASS-Transformer forward-projects the refined hyperspectral estimate through the CTIS imaging operator, compares the predicted sensor image with the measured image, and backprojects the residual to update the reconstruction. Through an unrolled network consisting of these two stages, PASS-Transformer enables physically faithful hyperspectral reconstruction with high fidelity. The main contributions of this work can be summarized as follows:

- An optimized low cost hyperspectral imaging system is built to effectively capture diffraction grating image. And a view-limited CTIS hyperspectral image reconstruction with only zero, first, and diagonal second diffraction orders is decomposed to a combination of a physics-based backprojection process and a transformer-based missing information inference;
- The novel Physics-Aware Spatial-Spectral Transformer (PASS-Transformer) is proposed to embed physics-based point-spread function operation and wavelength-dependent diffraction geometry directly into spatial and spectral attention mechanisms, respectively, leading to physics-constrained transformer learning;
- By using the forward and backward projections, my self-supervised reconstruction framework learns to iteratively recover physically plausible spatial-spectral information for hyperspectral cubes by enforcing forward consistency with the CTIS imaging physics, which also enables reconstruction without ground-truth spectral labels;
- The developed method significantly outperforms conventional EM-based reconstruction and unconstrained deep networks under extreme view limitations, and achieves improved spectral fidelity, spatial resolution, and time performance. Many real-world applications are demonstrated by the developed hardware and software system.

2 Related Work

Since my work focuses on hyperspectral image acquisition and its physics-aware deep learning reconstruction, only closely related work in imaging hardware and inversion are reviewed in this section.

2.1 CTIS Image Formation and Hyperspectral Imaging

The snapshot image formation model in mask-based systems [Arce et al., 2014, Gehm et al., 2007] relies heavily on the pseudo-random modulation introduced by the physical mask, which dictates the compressed sensing matrix. In contrast, the CTIS forward operator is rather deterministic. The system's behavior is governed strictly by the diffraction dispersion and the spatially varying Point Spread Function generated by the optical design. Therefore, my work focuses on CTIS. It is capable of reconstructing a three-dimensional data cube from a single two-dimensional detector exposure. Early foundational work by Descour and Dereniak [Descour and Dereniak, 1995] established the theoretical framework for CTIS using diffraction optics to generate multiple spectrally dispersed projections of the scene. Subsequent experimental demonstrations and system refinements further optimized the designs [Wagadarikar et al., 2008]. As hyperspectral imaging expands into practical, field-deployable applications, there is a growing push toward low-cost hardware architectures [Monakhova et al., 2020]. While these budget-friendly diffraction systems broaden accessibility, they inherently trade off ideal optical precision, requiring computational models to compensate the image formation degradation at the hardware level [Li et al., 2022, Wang et al., 2024]. While CTIS offers true snapshot capability, its reconstruction problem is ill-posed. Practical systems are often constrained to a few

diffraction orders due to detector size, signal-to-noise limitations, and optical efficiency. These constraints lead to severe view limitation in the forward operator, which makes accurate reconstruction challenging.

2.2 Statistical Model-based Methods for Hyperspectral Image Reconstruction

Traditional CTIS reconstruction methods rely on iterative optimization algorithms that explicitly model the system physics. Expectation–maximization (EM) approaches have been widely adopted due to their ability to incorporate Poisson noise statistics and non-negativity constraints [Lange and Carson, 1984, Descour and Dereniak, 1995]. Variants of maximum likelihood and maximum a posteriori estimation incorporating smoothness or sparsity priors have also been explored [Harmany et al., 2012]. Total Variation (TV) regularization [Yuan, 2016] formulates the reconstruction as a regularized least-squares optimization problem to balance the data fidelity with structural priors. Although model-based approaches achieve physically interpretable solutions, they suffer from several limitations. First, wavelength-dependent PSF and dispersion effects are often simplified, leading to spectral cross-talk and spatial artifacts in practice. Second, regularization alone cannot recover information lying in the CTIS operator under extreme view limitation. Third, convergence is slow and computationally expensive, particularly for high-resolution image cubes.

2.3 Learning-based Inverse Computing for Hyperspectral Image Reconstruction

Recent advances in deep learning have inspired data-driven approaches to hyperspectral reconstruction. Convolutional neural networks (CNNs) have been applied to reconstruct hyperspectral images from compressive or coded measurements [Miao et al., 2019, Han et al., 2025]. Transformer-based architectures have further demonstrated strong capability in modeling long-range dependencies in spectral–spatial data [Zhang and Abdulla, 2026]. Vision transformers [Dosovitskiy et al., 2021] have emerged as a dominant methodology in this space, providing unprecedented ability to model the complex, long-range dependencies required to accurately resolve high-dimensional data cubes from limited projections. Despite their empirical success, most learning-based approaches treat the imaging system as a black box and do not explicitly encode optical physics. As a result, these methods often require large training datasets, generalize poorly across different system configurations, and may violate physical consistency when applied outside the training distribution.

To address the limitations of purely data-driven models, physics-informed learning has emerged as a promising paradigm for solving inverse problems. Ongie *et al.* [Ongie et al., 2020] provided a comprehensive overview of integrating physical models with deep learning. Adler and Öktem [Adler and Öktem, 2018] proposed learned primal-dual frameworks that embed forward operators into iterative neural networks. In optical imaging, physics-informed approaches have been applied to phase retrieval, ptychography, and compressive imaging. Lin *et al.* [Lin et al., 2025] introduced a zero-order diffraction-guided spectral-wise transformer that incorporates optical calibration information into the reconstruction framework. But it relies heavily on network attention weights to learn dispersion compensation instead of explicit optical physics. Existing methods typically embed physics at the level of the forward model or loss function, rather than within the internal representation and attention mechanisms of modern architectures.

2.4 Differentiation from Prior Work

My proposed method differs fundamentally from existing CTIS reconstruction approaches in both formulation and architectural design. Instead of learning a direct inverse mapping or iterating a model-based solver, I decompose reconstruction into a physics-based backprojection and a learned spatial–spectral refinement module. In particular, the PSF and diffraction processes of the CTIS system are embedded directly into the transformer attention mechanisms, constraining token interactions to physically meaningful spatial and spectral couplings, while the physical consistency enforced by forward and backward projections iteratively reduces disagreement between the measured sensor image and the forward projection of the reconstructed hyperspectral cube. To the best of my knowledge, this is the first work to integrate wavelength-dependent PSF and diffraction constraints into transformer-based attentions for hyperspectral reconstruction within an unrolled reconstruction framework. This design enables data-efficient learning, improved generalization, and physically consistent reconstructions under extreme view limitation. Table 1 summarizes the key distinctions between the proposed method and representative prior approaches.

Table 1: Comparison with prior CTIS reconstruction methods.

Method	PSF and Diffraction -aware	Physics-consistent	Learning
EM-based CTIS	✓	✓	×
TV-regularized inversion	✓	Partial	×
CNN-based reconstruction	×	×	✓
Generic Transformer	×	×	✓
Proposed method	✓	✓	✓

3 A Low-Cost Snapshot Computed Spectral Imaging System

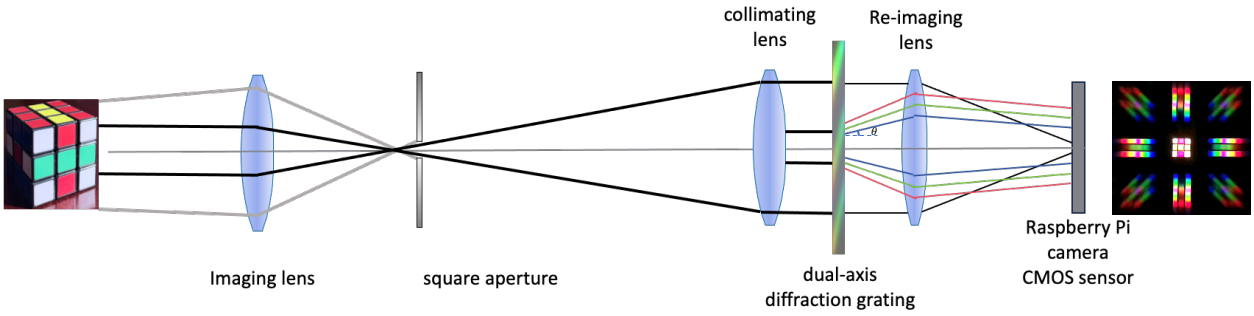
Since the CTIS image formation process relies on the strict mathematical integration of continuous PSFs across different wavelengths, it is well suited for my hardware-software co-design idea, leveraging the synergy between custom diffraction optics and physics-based computational reconstruction.

3.1 CTIS Imaging System Configuration

The most significant limitation of any CTIS system is that all information at the both spatial (the 2D image) and spectral (the 2D rainbow) domains must be captured on a single 2D image sensor. Because the raw data captured by the sensor is a "superposition" of spatial and spectral data, it cannot be used immediately. To prevent "spectral smear" from overlapping, the spatial resolution is drastically reduced. Inspired by the prior work [Habel et al., 2012, Salazar-Vazquez and Mendez-Vazquez, 2020], I develop a low-cost spectral imaging system based on the principles of CTIS using off-the-shelf parts, which disperse the zero, first, and diagonal second orders separately. As shown in Figure 1, in my developed CTIS system, a lens with $35mm$ and $1.6f$ is used as object lens, a 3D printed field stop aperture is placed before the collimating lens. The dual-axis diffraction grating (13,500 lines per inch) is used to disperse the incident light through



(a)



(b)

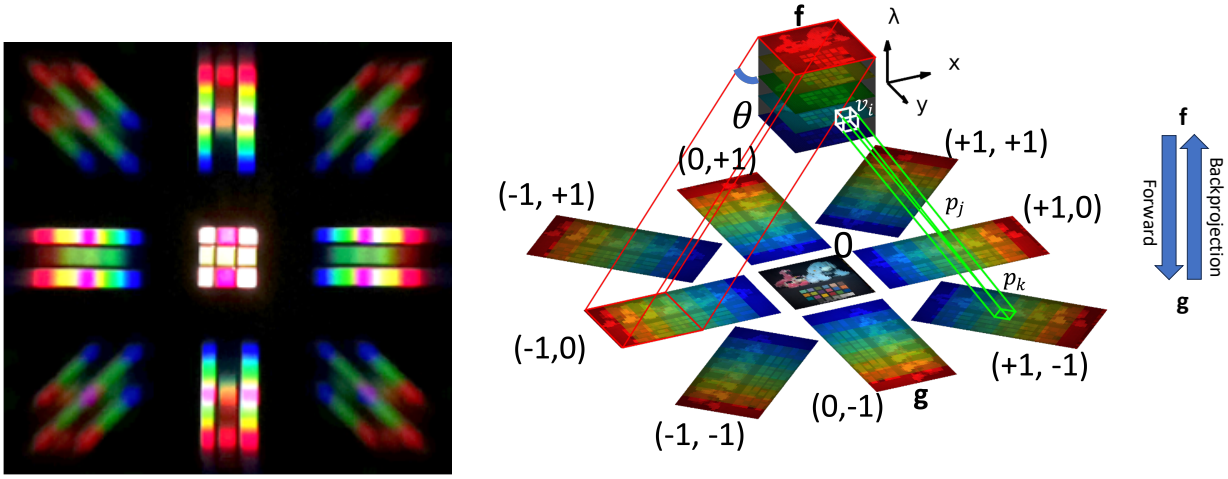
Figure 1: My low cost and high efficiency hyperspectral camera system. (a) An assembled prototype with a 3D-printed housing; (b) The illustration of the internal optical configuration.

a Raspberry Pi camera lens and re-imaged to its CMOS sensor. The total cost of these components is less than 300 dollars.

Based on these specific parameters, the "red tail" of the first order actually extends beyond the "blue head" of the second order. However, the first-order mode has a much stronger intensity than higher modes, making it the primary Region of Interest (ROI). The calibration process identifies the ROI areas to explicitly isolate the zero, first, and diagonal second order information (as shown in Figure 1(b)) and ignore other overlapping higher-order data during the reconstruction of the wavebands. The spectra can be recovered by solving a complex, under-determined linear system from these considered orders.

3.2 CTIS Forward Model with Zero, First, and Diagonal Second Orders

A fundamental property of diffraction gratings is that the intensity of the light decreases significantly as the order increases. In this work, I use three low-order ones, as denoted by their 2D diffraction orders $\mathbf{m} = (m_x, m_y)$, where $m_x, m_y \in \{-1, 0, 1\}$. As shown in Figure 2(a), these include zero-order $(0, 0)$, which contains a large portion of the total incident energy in the center of the image; first-order (i.e., $(1, 0), (-1, 0), (0, 1), (0, -1)$), which possesses significantly stronger intensity than other higher orders; and diagonal second-order (i.e., $(1, 1), (1, -1), (-1, 1), (-1, -1)$), which are the result of light being diffracted simultaneously by both axes of the grating, appearing at 45 degree angles relative to the primary axes. The diagonal second-order mode have useful intensity than any other higher-order modes. Because higher modes have significantly lower intensity, using them for reconstruction would require much longer exposure times



(a) (b)

Figure 2: (a) My CTIS camera sensored projection with the zero, first, and diagonal second order modes. (b) The illustration of the CTIS principle, which projects samples, such as V_i in the hyperspectral image cube, through wavelength defined dispersion angles with respect to the zero, first, and diagonal second order. The projected samples will be further blurred through point-spread function on the sensor image plane.

or more expensive, highly sensitive sensors. In my practical low-cost design, the signal-to-noise ratio in higher orders is often too poor to be useful. Including noisy data in the reconstruction algorithm often degrades the final quality of the hyperspectral cube rather than improving it. Therefore, except the diagonal second-order, the other higher orders are excluded during the hyperspectral cube reconstruction.

Assuming that the CTIS operates under only these modes (i.e., $m_x, m_y \in \{-1, 0, 1\}$), and spectral dispersion occurs along both spatial axes, as shown in Figure 2(b). The goal is to reconstruct a hyperspectral data cube, $f(x, y, \lambda) \in \mathbb{R}^{H\times W\times B}$, from these 2D multiplexed detector images corresponding to the diffraction orders. Let $g^{(\mathbf{m})}(x, y)$ denote the detector measurement (i.e., the image) for diffraction order $\mathbf{m} = (m_x, m_y)$. The forward imaging model is represented as follows,

$$g^{(\mathbf{m})}(x, y) = \sum_{k=1}^{B} \Big[f(x - m_x\alpha_x\lambda_k,\ y - m_y\alpha_y\lambda_k,\ \lambda_k) * h^{(\mathbf{m})}(x, y, \lambda_k) \Big] \tag{1}$$

where (α_x, α_y) are the spectral dispersion slopes, $h^{(\mathbf{m})}$ is the wavelength- and order-dependent point spread function, and $*$ denotes spatial convolution. Each detector pixel multiplexes contributions from multiple spatial locations and wavelengths, shifted by 2D dispersion and convolved by the PSF, i.e., multiple samples at $(x - m\alpha_x\lambda_k, y - m\alpha_y\lambda_k, \lambda_k)$ in the hyperspectral cube, f, contribute to a single location, (x, y), on the detector image g. The detector response, diffraction efficiency, and spectral sensitivity are also absorbed into the PSF to simplify the form of Equation 1.

This forward model can be written compactly in operator form as

$$\mathbf{g} = \mathbf{A}\mathbf{f}, \tag{2}$$

where $\mathbf{f} \in \mathbb{R}^{H \times W \times B}$ denotes the hyperspectral image cube, H and W are the spatial dimensions and B is the number of spectral bands. A CTIS system records a multiplexed sensor measurement $\mathbf{g}$ related to the hyperspectral cube through a known linear forward operator $\mathbf{A}$, which models wavelength-dependent dispersion, point-spread function blur, and summation on the detector. Mathematically, $\mathbf{A}$ is a many-to-one mapping due to the extreme sparsity of projection views. The inverse problem is severely ill-posed and admits a large null space. Therefore, given a g at (x, y), to solve f in the hyperspectral cube is severely under-determined.

3.3 Calibration of CTIS Parameters

To ensure the physical consistency of the reconstruction, the hyperspectral imaging system operator parameters, particularly the dispersion slopes and the point spread function, may be accurately determined through a pre-calibration process.

The dispersion slopes (α_x, α_y) define the spatial shift of spectral components on the 2D sensor. The diffraction geometry and resulting slopes are analytically computed based on the physical specifications of the diffraction grating period d and the optical focal length f. To determine how a wavelength λ at position (x, y) is mapped to the sensor, the zero-order mode is used as a spatial anchor. By using a Fluorescent Lamp with known emission peaks (e.g., $405nm$, $545nm$), my calibration algorithm calculates the displacement vector from the center.

The PSF, denoted as $h^{(\mathbf{m})}(x, y, \lambda_k)$, characterizes the system's impulse response for a specific diffraction order $\mathbf{m}$ and wavelength λ_k. The PSF is measured by scanning a monochromatic point source across the field of view for each target wavelength λ_k. Measurements are recorded for multiple wavelengths and each diffraction order, and $h^{(\mathbf{m})}(x, y, \lambda_k)$ are then obtained. The adjoint PSF, $h^{(\mathbf{m})\top}$, used in the backprojection initialization, is derived by applying a spatial kernel flipping operation to the calibrated PSF. The PSF is also weighted by the sensor's spectral response. Using an Incandescent Lamp (continuous spectrum), the spectral sensitivity S for each color channel (R,G,B) is calibrated, $H_c(\lambda) = \frac{I_{measured}(\lambda)}{I_{source}(\lambda)}$.

The $\mathbf{A}$, containing dispersion and PSF blur, is physically fixed once the device is assembled. Because the diffraction grating and the lens are in a static position relative to the sensor, it must be calibrated for each individual device to account for mechanical tolerances in 3D printing and assembly. To avoid the impracticality of independently measuring the system response at every spatial and spectral coordinate, a shift-invariant approximation is adopted to simplify the calibration process [Johnson et al., 2004]. For a CTIS hyperspectral imaging system, the discrete forward model based on the calibrated dispersion slope α and PSF h is shown in Equation 1 or compactly in Equation 2. The sensor sensitivity and wavelength are calibrated similarly as in [Habel et al., 2012, Salazar-Vazquez and Mendez-Vazquez, 2020].

4 Physics-Aware Spatial-Spectral Transformer for Self-Supervised Hyperspectral Image Cube Reconstruction

Following this forward mapping equation as shown in Equation 2, the 2D diffracted image on the sensor, $\mathbf{g}$, is projected from the 3D hyperspectral cube $\mathbf{f}$. To resort to machine learning to learn the inversion for

hyperspectral image reconstruction, the designed network must implicitly undo spatial and spectral blur, and the learning model is better to aware the image formation physics and facilitate better generalization with fewer samples.

4.1 Problem Definition for Physics-aware Hyperspectral Image Cube Reconstruction

Given a CTIS detected image, $\mathbf{g}$, the goal of the inversion is to solve, $\mathbf{f}$. Rather than learning a direct black-box inverse, I adopt the divide and conquer strategy and propose to decompose this CTIS inversion into a physics-based initialization component and a learned spatial–spectral refinement, as follows:

$$\tilde{\mathbf{f}} = \mathbf{A}^{\top}\mathbf{g} \; + \; \mathcal{N}_{\theta}(\mathbf{g}, \mathbf{A}), \tag{3}$$

where the adjoint backprojection $\mathbf{A}^{\top}\mathbf{g}$ provides a physics-based initialization that incorporates the calibrated CTIS imaging geometry and $\mathcal{N}_{\theta}$ is a learnable spatial–spectral refinement network parameterized by θ. This formulation explicitly separates the physics-based initialization from the learned correction in order to improve stability and accuracy. The spatial–spectral refinement focuses to learn additional priors from CTIS data to produce a structured correction that complements the physics-based backprojection. It does not attempt to recover unique full spectral content, but instead learns physically plausible spatial–spectral information consistent with PSF- and dispersion-induced coupling.

I resort to a transformer network to facilitate the spatial–spectral refinement. Therefore,

$$\tilde{\mathbf{f}} = \mathbf{A}^{\top}\mathbf{g} + \Phi_{\theta}(\mathbf{g}), \tag{4}$$

where Φ_{θ} is a transformer network trained to predict missing spatial–spectral information. It aims to resolve reconstruction ambiguities caused by overlapping spectral components and incomplete CTIS measurements, including fine-scale spectral variations and spatial-spectral mixing induced by dispersion. Equation 4 provides a conceptual decomposition, while the iterative optimization applies the transformer to the current hyperspectral estimate f_k, as defined in Equation 6.

4.2 Physics-Aware Spatial-Spectral Transformer

The CTIS inverse problem is decomposed naturally into an initialization component determined by the forward operator and a high-dimensional learned space, where my designed Physics-Aware Spatial-Spectral Transformer (PASS-Transformer) learns physically and statistically plausible spatial–spectral refinements while promoting measurement consistency. PASS-Transformer is uniquely designed to constrain PSF- and dispersion-attention and token mixing according to physical laws so that the transformer only aggregates information along physically motivated spatial–spectral coupling. The physical consistency constraint between the forward and back projection encourages the reconstruction to agree with the sensor measurement.

Specifically, my PASS-Transformer is composed of L identical blocks as shown in Figure 3. Each block consists of spatial attention constrained by the PSF and spectral attention guided by the dispersion. The PASS-Transformer network starts with a physics backprojection initialization, in which an explicit adjoint

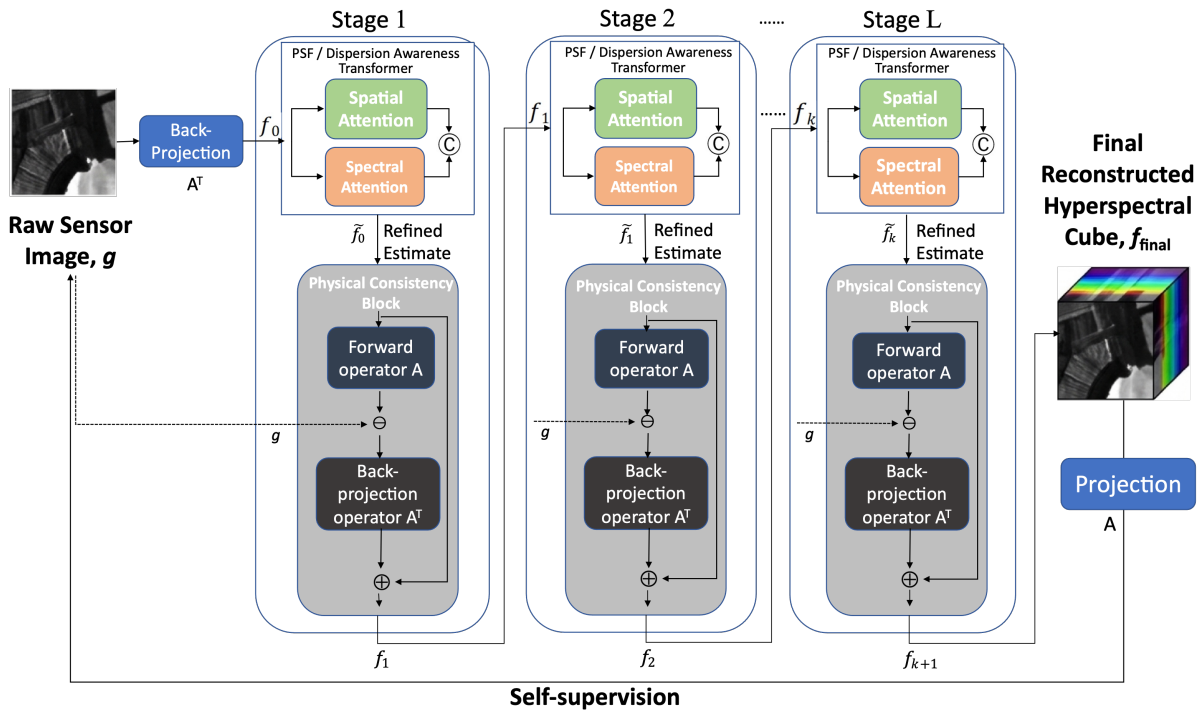


Figure 3: The physics-aware spatial-spectral deep unrolling network with self-supervision.

backprojection $\mathbf{A}^\top$ is applied to map the 2D snapshot measurement $\mathbf{g}$ into a physically valid hyperspectral space. This initialization utilizes the calibrated dispersion and PSF as described in Section 3.3 to ensure the initial 3D estimate $\mathbf{f}_0$ is grounded in the system's optical physics. During the iterative stages, the reconstruction is formulated as a deep unrolling network [Monga et al., 2021] that alternates between the learned refinement and the physical consistency. Unlike standard self-attention, the PSF-constrained spatial attention restricts token interactions to the effective PSF support. It ensures that tokens only attend to spatial locations that are physically coupled through the optical system, improving data efficiency and interpretability. With dispersion-aware spectral attention, this mechanism accounts for wavelength-dependent spatial mixing by modulating attention weights between spectral bands based on the system's dispersion slopes and PSF width. The transformer component is specifically trained to predict corrections that address spatial-spectral ambiguities in the view-limited CTIS reconstruction problem. I initialize the hyperspectral cube using the adjoint of the calibrated forward operator $\mathbf{A}^\top$ as follows.

$$\mathbf{f}_0 = \mathbf{A}^\top \mathbf{g}. \tag{5}$$

This step provides a meaningful initialization for the learning module. During the iterations,

$$\tilde{\mathbf{f}}_k = \mathbf{f}_k + \Phi_\theta(\mathbf{f}_k), \tag{6}$$

where $k \in \{0, 1, \cdots, L-1\}$, $\mathbf{f}_k$ is the input to the transformer and $\tilde{\mathbf{f}}_k$ is the transformer refined estimation.

Identity tokenization. Instead of projecting the hyperspectral cube into an abstract latent space, I represent each spatial pixel directly as a token. Specifically, the hyperspectral cube is reshaped into a set of $N = H \times W$ tokens, where H and W are the spatial dimensions and each token contains the full spectral vector with B-dimensional spectral vector $x_i \in \mathbb{R}^B$. This tokenization is a bijective reshaping operation and therefore preserves all information in the original cube. As a result, the token representation is mathematically equivalent to the hyperspectral cube itself, expressed in vectorized spatial form. No learned embedding or decoder is required. While identity tokenization slightly reduces representational flexibility compared to learned embeddings, it preserves exact physical correspondence between the transformer representation and the hyperspectral cube, enabling stable self-supervised training and precise enforcement of data consistency.

Spatial and spectral attention. The transformer operates on this tokenized representation using physics-aware attention mechanisms. Spatial self-attention models interactions between different spatial locations and is constrained by the calibrated point-spread function so that tokens only attend to physically coupled neighbors (see Section 4.3 for detail). Spectral self-attention operates within each token across the spectral channels, enabling band-to-band interactions that are further constrained by wavelength-dependent dispersion (see Section 4.4 for detail). Together, these attention mechanisms act as a learned spatial-spectral prior directly in the physical hyperspectral domain. Comparing the full 3D transformer with individually sampled spatial and spectral, my method saves more computation cost with factorized attentions.

Unrolled reconstruction with physical data consistency. The network is formulated as an unrolled iterative reconstruction. At each iteration, the transformer refines the current hyperspectral cube estimate through both spatial and spectral attentions. The refined estimate is then passed through a data consistency block, which enforces agreement with the measured sensor image by forward-projecting the estimate through the CTIS model and correcting it using the adjoint operator. Because the transformer output remains in the same physical space as the hyperspectral cube, data consistency can be enforced without any decoding step (see Section 4.5 for detail). The explicit data consistency step promotes that all learned refinements stay consistent with the sensor measurement, while the transformer captures physically plausible spatial-spectral redistributions that are not uniquely determined by the CTIS measurement.

4.3 PSF-Constrained Spatial Attention

Standard self-attention allows unrestricted token interactions. In contrast, I restrict spatial attention using the effective PSF support. The Transformer operates on spatial-spectral tokens indexed by spatial positions (x_i, y_i), where $i \in \{1, \ldots, HW\}$. Each token contains a spectral vector $\mathbf{f}_i \in \mathbb{R}^B$, defined as $\mathbf{f}_i = [f(x_i, y_i, \lambda_1), \ldots, f(x_i, y_i, \lambda_B)]^\mathsf{T}$, where H and W denote the spatial dimensions and B is the number of spectral bands. Attention weights are restricted using the PSF-induced coupling map such that tokens may only attend to other tokens whose forward projections overlap within the PSF support on the sensor as shown in Figure 4. For tokens i and j located at spatial coordinates $\mathbf{x}_i$ and $\mathbf{x}_j$, the attention weight is defined

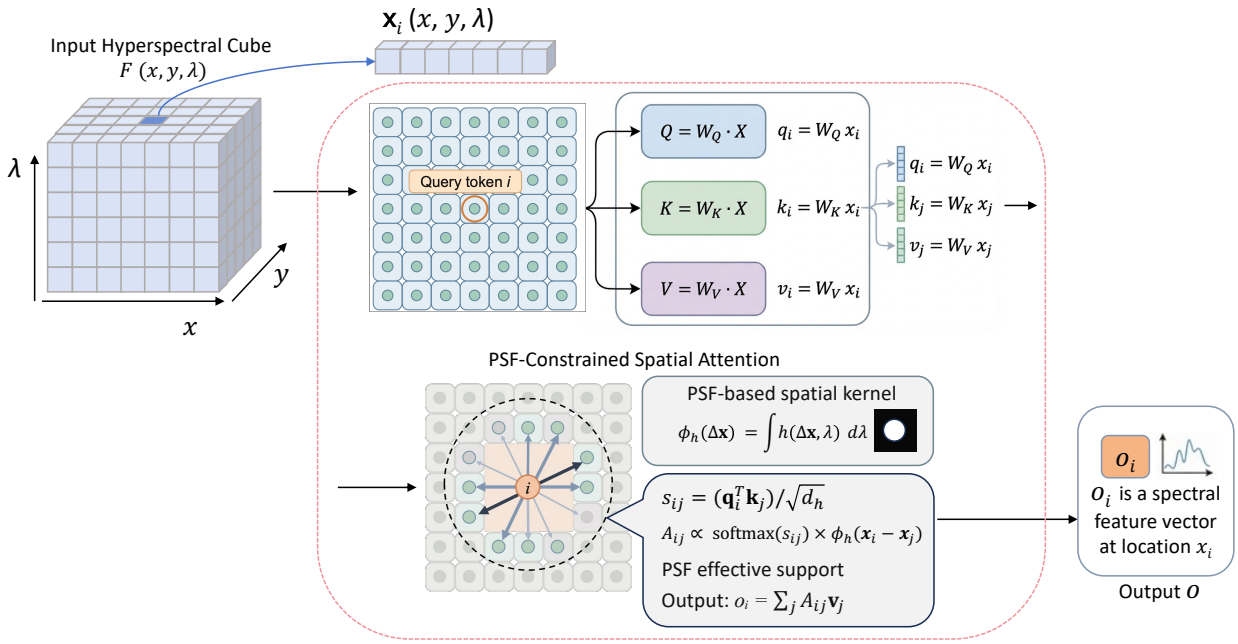


Figure 4: The proposed PSF-constrained spatial attention.

as

$$A_{ij} = \frac{\exp\left(\frac{Q_i K_j^\top}{\sqrt{d_h}}\right) \cdot \phi_h(\mathbf{x}_i - \mathbf{x}_j)}{\sum_{j'} \exp\left(\frac{Q_i K_{j'}^\top}{\sqrt{d_h}}\right) \cdot \phi_h(\mathbf{x}_i - \mathbf{x}_{j'})}, \tag{7}$$

where Q_i and K_j are the query and key vectors, d_h is the embedding dimension d over the number of attention heads h, and ϕ_h is a PSF-derived weighting function

$$\phi_h(\Delta\mathbf{x}) = \int h(\Delta\mathbf{x}, \lambda)\, d\lambda, \tag{8}$$

where $h(\Delta x, \lambda)$ denotes an effective PSF obtained by aggregating the calibrated order-dependent PSFs $h^{(m)}$ across the nine retained diffraction orders. This enforces that tokens only attend to spatial locations derived from calibrated optical system, as shown in Figure 4. Note that, this is not only a learned map but also derived from optics with effective point spread function support. Therefore, this attention mechanism incorporates a calibrated optical prior into the learned spatial interactions, as compared to the blackbox learned deblurring, which often introduces hallucinated structures and demands lots of training data.

4.4 Spectral Attention with Dispersion Awareness

Spectral attention is applied within each spatial location. Within each spatial token, the B spectral bands serve as individual attention positions, with learned projections generating band-wise query, key, and value features to form a $B \times B$ spectral attention matrix. To account for wavelength-dependent spatial mixing,

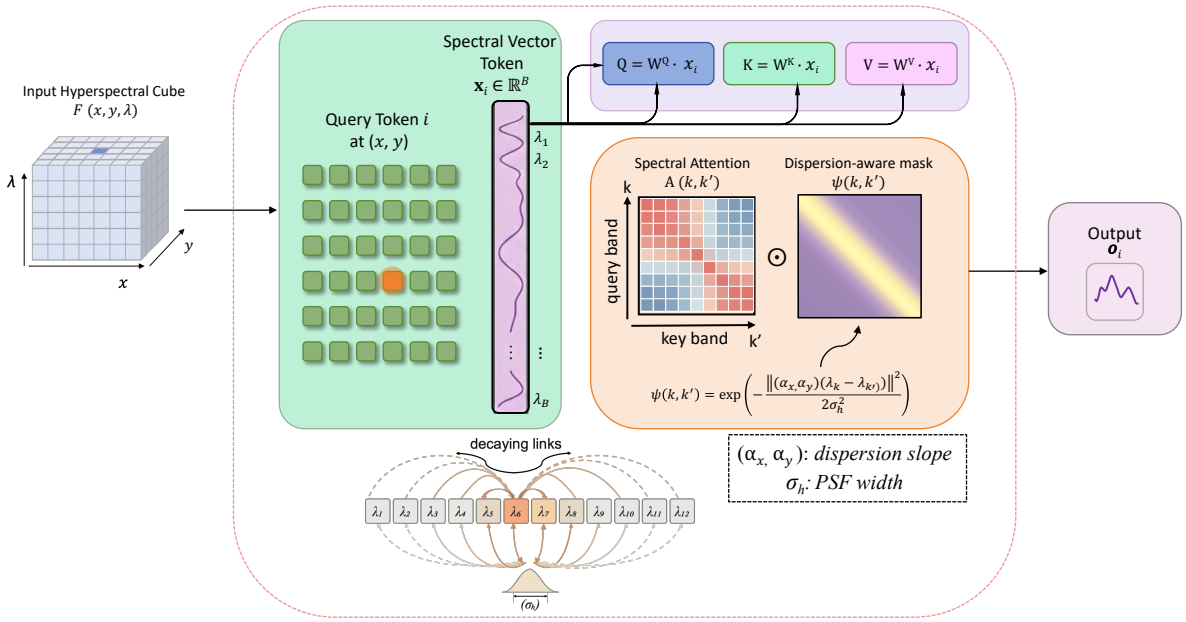


Figure 5: The proposed dispersion-constrained spectral attention.

attention weights between spectral bands k and k' are modulated as

$$\psi(k, k') = \exp\left(-\frac{\|(\alpha_x, \alpha_y)(\lambda_k - \lambda_{k'})\|^2}{2\sigma_h^2}\right), \tag{9}$$

where σ_h characterizes the PSF width. This equation calculates how much cross-talk or interaction exist between different bands based on their physical overlap on the detector. Unlike standard transformers that learn relationships purely from data (a "black box"), this mechanism embeds the physical spectral cross-talk induced by dispersion directly into the attention mechanism. If the two wavelengths are dispersed far apart on the imaging sensor, $\psi(k, k')$ becomes very small. They should not attend to each other strongly. It restricts the "search space" of the attention mechanism to physically valid light paths as shown in Figure 5. Spectral attention follows Equation 7, replacing the PSF-derived spatial kernel with the dispersion-aware kernel $\psi(k, k')$, followed by normalization of the attention weights. This also makes the model more interpretable and computationally efficient compared to unconstrained deep networks.

To encourage transformer representations to align with physical laws, spatial attention is constrained by a PSF-derived spatial kernel, Equation 8, whereas spectral attention is modulated by a dispersion-aware mask, Equation 9. Through identity tokenization, the spatial and spectral masks modulate attention weights computed from the query Q and key K tensors, guiding the aggregation of value V features according to physically motivated spatial and spectral relationships.

4.5 Physical Consistency in the Forward and Inverse Mapping

Physical Consistency (PC) Block is used to enforce measurement fidelity. It simulates the measurement process using the forward operator $\mathbf{A}$, calculates the residual error against the input $\mathbf{g}$, and applies a corrective

update to ensure that the reconstructed hyperspectral cube remains physically aligned by enforcing agreement with the original sensor measurement. More specifically, the block takes the refined hyperspectral cube estimate from the transformer and passes it through the forward operator $\mathbf{A}$. This process simulates what the sensor would see if the current estimate were the true scene. The network computes the residual error by subtracting the simulated sensor image from the raw sensor image $\mathbf{g}$. This residual error in 2D sensor space is then mapped back into the 3D hyperspectral domain using the backprojection operator $\mathbf{A}^\top$, which is the physical adjoint of the forward model. The mapped error is added back to the current estimate to produce the updated 3D estimate $\mathbf{f}(k+1)$. This encourages the learned corrections from the transformer to better agree with the measured sensor image.

The procedure is derived from the optimization of the physical consistency objective function. Mathematically, let the physical consistency term be

$$L_{\mathrm{PC}}(\tilde{\mathbf{f}}_k) = \frac{1}{2}\|\mathbf{g} - \mathbf{A}\tilde{\mathbf{f}}_k\|^2, \tag{10}$$

To derive the update used in the physical consistency block, I minimize the data-fidelity term in Equation 10. Taking the gradient with respect to $\tilde{\mathbf{f}}_k$ yields

$$\nabla_{\tilde{\mathbf{f}}_k} L_{\mathrm{PC}}(\tilde{\mathbf{f}}_k) = \mathbf{A}^\top \mathbf{A}\tilde{\mathbf{f}}_k - \mathbf{A}^\top \mathbf{g} = \mathbf{A}^\top(\mathbf{A}\tilde{f}_k - \mathbf{g}). \tag{11}$$

Therefore, a gradient-descent step for enforcing physical consistency is

$$\mathbf{f}_{k+1} = \tilde{\mathbf{f}}_k - \eta_k \mathbf{A}^\top(\mathbf{A}\tilde{\mathbf{f}}_k - \mathbf{g}). \tag{12}$$

where η_k is the optimization step size. This shows that the physical consistency block corrects the current estimate $\tilde{\mathbf{f}}_k$ by first computing the measurement residual $\mathbf{g} - \mathbf{A}\tilde{\mathbf{f}}_k$ in sensor space, and then backprojecting that residual through the adjoint operator $\mathbf{A}^\top$ into hyperspectral cube space, as shown in the Physical Consistency Block in Figure 3. Thus, the physical consistency block is a physics-guided residual-correction step that drives the reconstruction toward agreement with the measured data under the forward model $\mathbf{g} = \mathbf{A}\mathbf{f}$.

4.6 Self-Supervised Loss Function

The self-supervised network is trained end-to-end using a composite loss

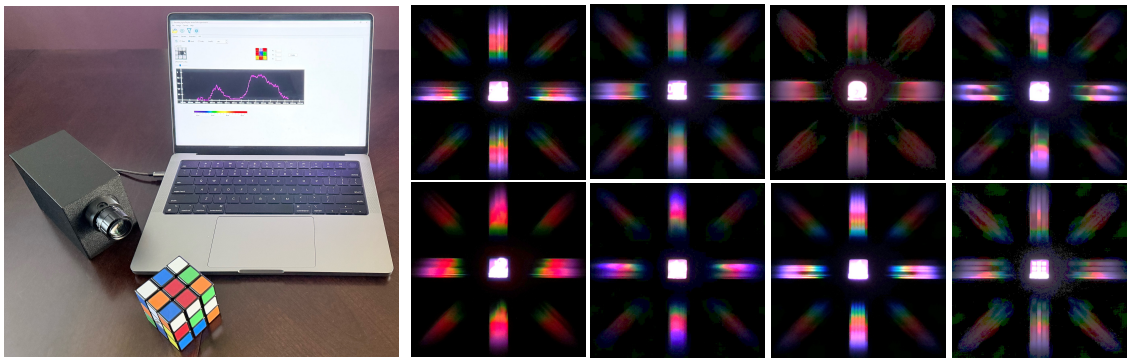
$$\mathcal{L} = \|\mathbf{A}\mathbf{f} - \mathbf{g}\|_1 + \beta\|\nabla_\lambda \mathbf{f}\|_1 + \gamma\|\nabla_{x,y}\mathbf{f}\|_1, \tag{13}$$

where the terms respectively enforce physical forward consistency, spectral smoothness, and spatial regularity using the final physical consistency-updated result, $\mathbf{f} = \mathbf{f}_L$. By leveraging the known CTIS forward model, the proposed framework eliminates the need for paired sensor measurements and ground-truth hyperspectral cubes, which are expensive to acquire. As a result, the network can be trained in a fully self-supervised manner and reliably converges using only the CTIS acquired data.

5 Experiments

All CTIS input measurements used in this work are collected using the developed low-cost CTIS imaging system (as described in Section 3). The training is performed entirely using real measurements from the physical device and the calibrated CTIS forward operator.

5.1 Training Data Collection



(a) (b)

Figure 6: (a) The developed CTIS hardware and software system; (b) Some of the collected CTIS snapshot samples.

Each training sample begins with a single CTIS exposure containing the 9 segmented diffraction orders. A fixed region-of-interest (ROI) per order is cropped using the calibration-derived centers and dispersion geometry. Let the extracted order images $\{g^{(\mathbf{m})}\}_{\mathbf{m}\in\mathcal{M}}$ be the retained set of orders (zero, first, and diagonal second orders). The wavelength- and order-dependent PSFs $\{h^{(\mathbf{m})}(\lambda_k)\}$ and dispersion slopes (α_x, α_y) are obtained via the calibration procedure (as described in Section 3.3) and treated as fixed during training and inference. These parameters define the discrete forward operator $\mathbf{A}$ and its adjoint $\mathbf{A}^\top$.

I use the developed CTIS system and collect 500 images (some samples are shown in Figure 6(b)) from diverse small objects with different textures and materials under incandescent light. I split samples into train/val/test (i.e., 80/10/10) by capture session to avoid leakage across near-duplicate frames.

5.2 Implementation Details

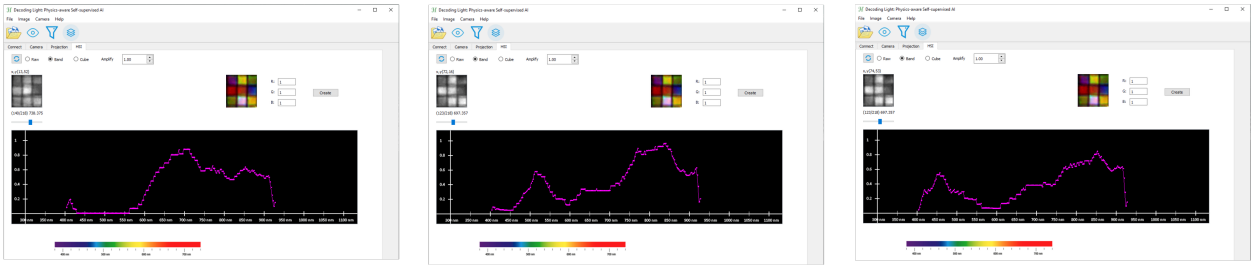
I implement PASS-Transformer in PyTorch. The PASS-Transformer processes a hyperspectral data cube with spatial dimensions $H = 100$, $W = 100$, and $B = 105$ spectral bands. The model uses $L = 6$ transformer iteration blocks with embedding dimension $d = B$. The architecture employs $h = 3$ attention heads within its physics-aware transformer blocks. Spatial attention is computed per pixel, and attention is masked/weighted using the calibrated PSF support. Spectral attention is applied per-pixel along the B spectral bands within a pixel, modulated by the dispersion-aware kernel. The self-supervised loss function uses a spectral smoothness weight of $\beta = 10^{-2}$ and a spatial regularity weight of $\gamma = 5\times10^{-3}$. The iteration

begins from the physics backprojection initialization $\mathbf{f}_0 = \mathbf{A}^\top \mathbf{g}$, followed by L unrolled refinement steps. At each iteration, physical consistency is enforced through Equation 12 using a learned step size η_k.

I train on a single NVIDIA GeForce RTX 3090 GPU for 20 epochs using Adam [Kingma and Ba, 2015], initial learning rate 10^{-4}, weight decay 10^{-5}, and cosine annealing. I use mixed precision training and gradient clipping to improve stability in early epochs. The batch size is 4, which is chosen to maximize GPU utilization.

5.3 Reconstruction Results and Comparison to State-of-the-Art Methods

A reconstruction example using the developed system is presented in Figure 7. The software allows interactive spectral evaluation of the reconstructed hyperspectral image cube. When a specific spatial location is selected via the single-band reference image (top left), the full continuous 1D spectral profile for that exact pixel is rendered at the bottom. This visualization demonstrates the network's capability to reliably recover distinct spectral signatures across the visible and near-infrared spectrum (400 nm to 920 nm) directly from the compressed 2D diffraction snapshot. The synthesized RGB image is displayed at the top right.



(a) (b) (c)

Figure 7: Reconstruction results using the developed system. Each panel illustrates a reference band (top left), a synthesized RGB image (top right), and the reconstructed continuous spectral profile (bottom). The extracted spectral profiles correspond to a selected (a) red pixel, (b) green pixel, and (c) blue pixel.

I compare to the representative baselines spanning physics-only inversion and learned reconstruction, including

- EM CTIS reconstruction (physics-only): iterative maximum-likelihood inversion with non-negativity.
- TV-regularized inversion: physics forward model along with total variation prior solved by iterative optimization.
- Blackbox Transformer: identical capacity to PASS-Transformer but without PSF/dispersion constrained attentions.
- Unrolled network without physics-aware attention: unrolling with standard attention modules.

The primary metric is sensor-domain physics consistency:

$$E_{\mathrm{DC}} = \frac{\|\mathbf{A}\mathbf{f}_L - \mathbf{g}\|_1}{\|\mathbf{g}\|_1}, \qquad E_{\mathrm{DC}}^{(2)} = \frac{\|\mathbf{A}\mathbf{f}_L - \mathbf{g}\|_2}{\|\mathbf{g}\|_2}. \tag{14}$$

Method	$E_{\mathrm{DC}} \downarrow$	$E_{\mathrm{DC}}^{(2)} \downarrow$	SAM $\downarrow$	ms/frame$\downarrow$
EM [Descour and Dereniak, 1995]	0.180	0.120	0.310	2900
TV-regularized [Yuan, 2016]	0.140	0.095	0.260	1600
Blackbox Transformer [Vaswani et al., 2017]	0.115	0.078	0.205	34
Unrolled (no physics-attn)	0.108	0.072	0.185	36
PASS-Transformer (Mine)	**0.052**	**0.036**	**0.135**	**31**

Table 2: Comparison using sensor-domain physics consistency and spectral similarity.

Variant	$E_{\mathrm{DC}} \downarrow$	SAM $\downarrow$	ms/frame$\downarrow$
Full PASS-Transformer (Mine)	**0.052**	**0.135**	31
w/o PSF spatial constraint	0.066	0.165	30
w/o dispersion spectral constraint	0.063	0.175	30
w/o physical consistency block	0.102	0.195	28

Table 3: Ablation study isolating the impact of physics-aware attentions and physical data consistency.

To evaluate spectral accuracy, I compute the Spectral Angle Mapper (SAM) between reconstructed spectra and independently measured reference spectra from a calibrated spectrometer, aligned across the same B wavelength bands.

$$\mathrm{SAM}(\mathbf{s}, \hat{\mathbf{s}}) = \cos^{-1}\left(\frac{\mathbf{s} \cdot \hat{\mathbf{s}}}{\|\mathbf{s}\|_2 \, \|\hat{\mathbf{s}}\|_2}\right), \tag{15}$$

where $\mathbf{s}, \hat{\mathbf{s}} \in \mathbb{R}^B$ denote the reference and reconstructed spectral vectors, respectively. Table 2 reports the quantitative comparison on the test set along with the runtime performance in ms/frame. The best results are highlighted. In this experiment, my PASS-Transformer achieves a normalized sensor-domain L1 consistency error of 0.052, compared with 0.180 for EM reconstruction, representing a 71.1% reduction in measurement error. It also improves spectral fidelity and achieves real-time inference. Runtime is averaged per image using batch size 1 on the RTX 3090, excluding the visualization overhead.

5.4 Model Analysis and Ablation Study

I ablate PASS-Transformer components that encode physics knowledge, including PSF-constrained spatial attention, dispersion-aware spectral attention, and physical consistency block. All ablations keep total parameters approximately constant where possible. I train each variant with identical optimizer settings and evaluate on the same test set. The experiment shows the physics-aware attentions reduce spatial and spectral cross-talk and improves SAM, and the physical consistency block reduces sensor-domain error and improves accuracy. All the components contribute to the reconstruction accuracy as shown in Table 3.

6 System and Applications to Precision Agriculture, Environmental Engineering, and Food Ingredient Detection

The developed CTIS system with PASS-Transformer reconstruction can be applied to potential real-world deployments. It supports CTIS data acquisition, hyperspectral image reconstruction, spatial and spectral

visualization, and advanced quantitative analysis. In this section, I demonstrate the practical utility of the system across three representative application domains, including precision agriculture, environmental engineering, and food ingredient detection. All application demonstrations use the same hardware configuration and reconstruction pipeline, with no task-specific network retraining. This highlights the generality of the proposed framework. The reconstructed hyperspectral image cubes enable precise pixel- and region-level spectral inspection of visually indistinguishable materials. To analyze targeted areas, a region-level spectral profile can be extracted by averaging the spatial dimensions of the hyperspectral cube,

$$f_{\text{region}}(\lambda) = \frac{1}{hw} \sum_{x=1}^{w} \sum_{y=1}^{h} f(x, y, \lambda). \tag{16}$$

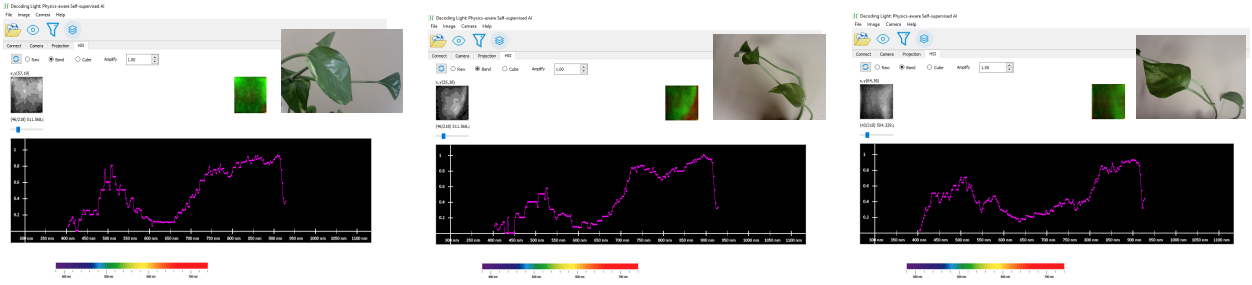
where h and w are the height and width of the region-of-interest.

6.1 Application I: Precision Agriculture

Precision agriculture relies on timely assessment of crop health, stress, and nutrient status. Traditional hyperspectral imaging systems are often too slow or costly for widespread deployment. Using my developed system, I image plant leaves and small plots under indoor lighting conditions. The reconstructed hyperspectral image cubes by PASS-Transformer enable precision monitoring of plant conditions. Specifically, the system analyzes critical spectral cues, e.g., Normalized Difference Vegetation Index (NDVI) contrast. The NDVI is calculated as follows,

$$\text{NDVI} = \frac{R_{800} - R_{680}}{R_{800} + R_{680}}, \tag{17}$$

which compares near-infrared (NIR) and red reflectance. Because healthy leaves reflect more NIR light and absorb more red light, they produce higher NDVI values. As the cases shown in Figure 8, the analysis identifies "Most healthy" samples with an NDVI of 0.71 (Figure 8(a)), standard "Healthy" samples at 0.58 (Figure 8(b)), and "Stressed" samples at 0.43 (Figure 8(c)). I exam a total of 20 samples, and this spectral inspection pipeline achieves a 95% accuracy rate in classifying healthy versus stressed plants.



(a) (b) (c)

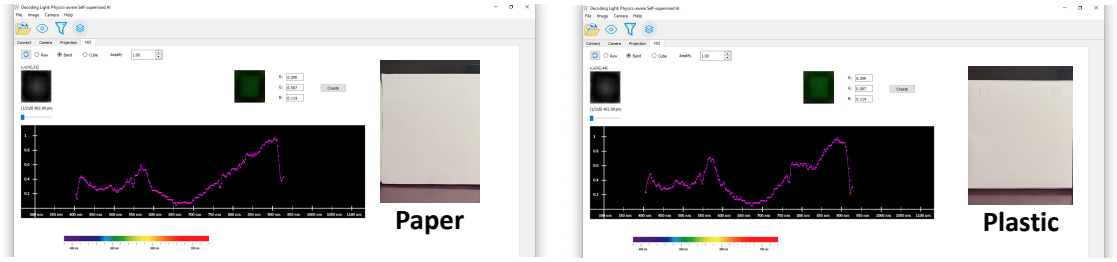
Figure 8: Application of the reconstructed hyperspectral imaging data for precision agriculture, demonstrating region-level spectral inspection of visually indistinguishable plant health states. The interface calculates NDVI, differentiating between healthy leaves (NDVI $= 0.71$ (a) to 0.58 (b)) and stressed (NDVI $= 0.43$ (c)).

6.2 Application II: Environmental Engineering

Environmental monitoring often requires identifying material composition or contamination. I apply the developed system to automatic recycling material classification between visually similar items such as paper and plastic of similar colors. The reconstructed hyperspectral image cubes can reveal material-dependent spectral behavior. Therefore, the classification can rely on the overall structure curvature of the spectral profile. To quantitatively compare these spectral signatures, the Spectral Angle Mapper (SAM) is utilized here as well,

$$\mathrm{SAM}(\mathbf{s}, \mathbf{r}) = \cos^{-1}\left(\frac{\mathbf{s} \cdot \mathbf{r}}{\|\mathbf{s}\|\|\mathbf{r}\|}\right). \tag{18}$$

To enable robust automated classification via the Spectral Angle Mapper, standard template spectral profiles, serving as the reference vectors r, are first established for a target material class, e.g., paper and plastic in this case. These templates are generated by capturing hyperspectral image cubes of pure, authenticated samples under controlled lighting conditions. By applying the previously defined region-level averaging formulation to uniform areas, a highly representative 1D spectral signature is extracted and stored for each material. During the inference phase, the spectral profile of an unknown sample, denoted as s, is extracted and compared against these templates. The system computes the SAM between the unknown signal s and each candidate reference r, and assigns the classification to certain material category that yields the minimum spectral angle, which represents the highest morphological similarity between the spectral curves. Note that, the reference template and evaluated samples are independent. By evaluating these distinct spectral profiles, the system achieves a 97% material classification accuracy in differentiating papers (50 samples) and plastics (50 samples) with similar appearances as shown in Figure 9.

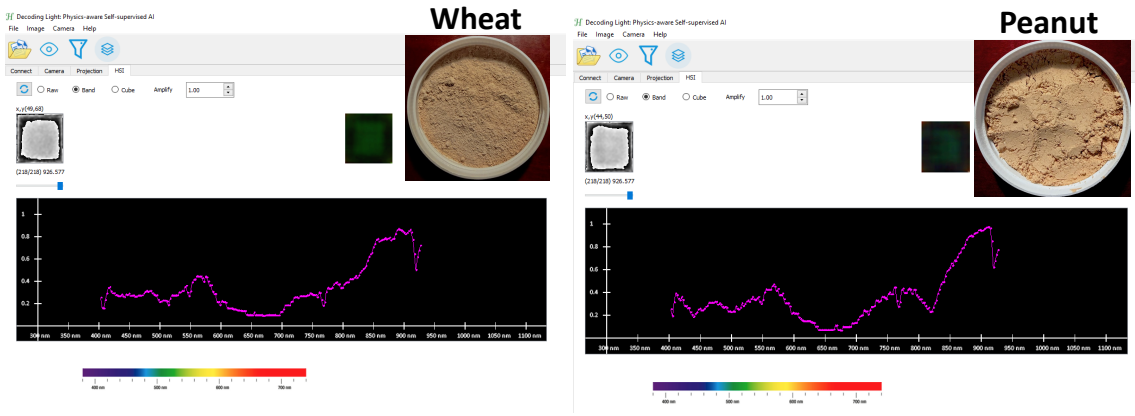


(a) (b)

Figure 9: Application in environmental engineering for automatic recycling material classification. The system utilizes SAM to distinguish between visually similar paper (a) and plastic (b) samples, achieving a 97% classification accuracy.

6.3 Application III: Food Ingredient Classification

Food allergies are often triggered by trace ingredients (e.g., nuts, dairy, soy) that may not be visually distinguishable. Many allergenic compounds exhibit characteristic reflectance or absorption patterns in the visible and near-infrared spectrum. In this application, the developed hyperspectral imaging system is employed for a preliminary food ingredient classification demonstration, particularly focusing on nut identification be-



(a) (b)

Figure 10: Application of the system for food ingredient classification, specifically distinguishing peanut (b) and wheat (a) powders using reconstructed spectral signatures. Relying on SAM, the method achieves a 95% nut classification accuracy.

tween visually indistinguishable powders like wheat and peanut. The spectral signatures are compared using the SAM formulation as illustrated in Equation 18. Similarly, representative spectral signatures are extracted and stored for wheat and peanut. Leveraging these spectral variances, the framework achieves a 95% nut classification accuracy among 40 testing samples based on SAM. In the experiment, the reference templates and evaluated samples are independent. Figure 10 illustrates reconstructed spectral profiles from visually similar samples. Even when RGB appearance is nearly identical, reconstructed spectra exhibit consistent differences in specific wavelength ranges corresponding to ingredient composition.

Across all three application domains, the developed CTIS and PASS-Transformer system demonstrate the ability to accurately extract and analyze spectral information from a single snapshot capture.

7 Conclusion

In this work, I have presented a novel physics-aware spatial-spectral transformer for a self-supervised hyperspectral imaging reconstruction system. By integrating PSF-constrained spatial attention, dispersion-aware spectral attention, and iterative physical data consistency, the proposed framework enables accurate and efficient self-supervised reconstruction without requiring ground-truth hyperspectral labels. This work further expands the accessibility of hyperspectral imaging and demonstrates its potential for diverse applications in precision agriculture, environmental monitoring, and food safety.

In the future, I plan to investigate deep learning-based approaches for higher-order modes during signal processing and image reconstruction, and hope that finer-scale spatial and spectral details can be enhanced in this system. I also plan to explore more applications on quantitative classification and detection with the imaging device and PASS-Transformer.

8 References

[Adler and Öktem, 2018] Adler, J. and Öktem, O. (2018). Learned primal-dual reconstruction. *IEEE Transactions on Medical Imaging*, 37(6):1322–1332.

[Arce et al., 2014] Arce, G. R., Brady, D. J., Carin, L., Arguello, H., and Kittle, D. S. (2014). Compressive coded aperture spectral imaging: An introduction. *IEEE Signal Processing Magazine*, 31(1):105–115.

[Bhargava et al., 2024] Bhargava, A., Sachdeva, A., Sharma, K., Alsharif, M. H., Uthansakul, P., and Uthansakul, M. (2024). Hyperspectral imaging and its applications: A review. *Heliyon*, 10(12):e33208.

[Descour and Dereniak, 1995] Descour, M. and Dereniak, E. (1995). Computed-tomography imaging spectrometer: experimental calibration and reconstruction results. *Applied Optics*, 34(22):4817–4826.

[Di et al., 2025] Di, J., Jin, Y., Yu, Z., Cheng, L., Ma, J., Zhan, N., and Xu, K. (2025). Spatial-enhanced reflective coded aperture snapshot spectral imaging. *arXiv preprint arXiv:2504.20516*.

[Dosovitskiy et al., 2021] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*.

[Gehm et al., 2007] Gehm, M. E., John, R., Brady, D. J., Willett, R. M., and Schulz, T. J. (2007). Single-shot compressive spectral imaging with a dual-disperser architecture. *Optics Express*, 15(21):14013–14027.

[Habel et al., 2012] Habel, R., Kudenov, M., and Wimmer, M. (2012). Practical spectral photography. *Computer Graphics Forum*, 31(2):449–458.

[Han et al., 2025] Han, X.-H., Wang, J., and Jiang, H. (2025). Recent advancements in hyperspectral image reconstruction from a compressive measurement. *Sensors*, 25(11):3286.

[Harmany et al., 2012] Harmany, Z. T., Marcia, R. F., and Willett, R. M. (2012). This is SPIRAL-TAP: Sparse Poisson Intensity Reconstruction ALgorithms—theory and practice. *IEEE Transactions on Image Processing*, 21(3):1084–1096.

[He et al., 2017] He, L., O'Sullivan, J., Politte, D., and Arvidson, R. (2017). Resolution analysis of regularized maximum log-likelihood reconstruction method for CRISM hyperspectral data. In *Imaging Systems and Applications*, page JTu5A.16.

[Johnson et al., 2004] Johnson, W. R., Hege, K., O'Connell, D., and Dereniak, E. L. (2004). Recovery of a flawed hyperspectral imager calibration using optical modeling. *Optics Express*, 12(10):2251–2257.

[Johnson et al., 2006] Johnson, W. R., Wilson, D. W., and Bearman, G. (2006). Spatial–spectral modulating snapshot hyperspectral imager. *Applied Optics*, 45(9):1898–1908.

[Kingma and Ba, 2015] Kingma, D. P. and Ba, J. (2015). Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*.

[Lange and Carson, 1984] Lange, K. and Carson, R. (1984). EM reconstruction algorithms for emission and transmission tomography. *J Comput Assist Tomogr*, 8(2):306–16.

[Li et al., 2022] Li, L., Wang, L., Song, W., Zhang, L., Xiong, Z., and Huang, H. (2022). Quantization-aware deep optics for diffractive snapshot hyperspectral imaging. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19780–19789.

[Lin et al., 2025] Lin, D., Song, Y., Wang, H., Ji, Y., Lai, J., and Li, Z. (2025). Zero-order diffraction-guided spectral-wise transformer: a lightweight and efficient framework for computed tomography imaging spectrometry. *Optics Express*, 33(21):44011–44025.

[Luo et al., 2021] Luo, L., Li, S., Yao, X., and He, S. (2021). Rotational hyperspectral scanner and related image reconstruction algorithm. *Scientific Reports*, 11(1):3296.

[Miao et al., 2019] Miao, X., Yuan, X., Pu, Y., and Athitsos, V. (2019). λ-net: Reconstruct hyperspectral images from a snapshot measurement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4058–4068.

[Monakhova et al., 2020] Monakhova, K., Yanny, K., Aggarwal, N., and Waller, L. (2020). Spectral DiffuserCam: lensless snapshot hyperspectral imaging with a spectral filter array. *Optica*, 7(10):1298–1307.

[Monga et al., 2021] Monga, V., Li, Y., and Eldar, Y. C. (2021). Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing. *IEEE Signal Processing Magazine*, 38(2):18–44.

[Ongie et al., 2020] Ongie, G., Jalal, A., Metzler, C., Baraniuk, R., Dimakis, A., Willett, R., and Soltanolkotabi, M. (2020). Deep learning techniques for inverse problems in imaging. *IEEE Journal on Selected Areas in Information Theory*, 1(1):39–56.

[Salazar-Vazquez and Mendez-Vazquez, 2020] Salazar-Vazquez, J. and Mendez-Vazquez, A. (2020). A plug-and-play hyperspectral imaging sensor using low-cost equipment. *HardwareX*, 7:e00087.

[Shi et al., 2024] Shi, Z., Dun, X., Wei, H., Dong, S., Wang, Z., Cheng, X., Heide, F., and Peng, Y. (2024). Learned multi-aperture color-coded optics for snapshot hyperspectral imaging. *ACM Transactions on Graphics*, 43(6):208.

[Vaswani et al., 2017] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*, pages 5998–6008.

[Wagadarikar et al., 2008] Wagadarikar, A. A., John, R., Willett, R. M., and Brady, D. J. (2008). Single disperser design for coded aperture snapshot spectral imaging. *Applied Optics*, 47(10):B44–B51.

[Wang et al., 2024] Wang, L., Li, L., Song, W., Zhang, L., Xiong, Z., and Huang, H. (2024). Non-serial quantization-aware deep optics for snapshot hyperspectral imaging. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(11):6993–7010.

[Wang et al., 2025] Wang, Z., Peng, Y., Fang, L., and Gao, L. (2025). Computational optical imaging: on the convergence of physical and digital layers. *Optica*, 12(1):113–130.

[Xie et al., 2022] Xie, S., Wang, S., Song, C., and Wang, X. (2022). Hyperspectral image reconstruction based on spatial-spectral domains low-rank sparse representation. *Remote Sensing*, 14(17):4184.

[Xiong et al., 2017] Xiong, Z., Wang, L., Li, H., Liu, D., and Wu, F. (2017). Snapshot hyperspectral light field imaging. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3270–3278.

[Yuan, 2016] Yuan, X. (2016). Generalized alternating projection based total variation minimization for compressive sensing. In *2016 IEEE International Conference on Image Processing*, pages 2539–2543.

[Yue et al., 2024] Yue, Q., Chen, B., Liu, X., and Zheng, Z. (2024). Snapshot spectral imaging based on aberration model-driven deep learning. *Optics Letters*, 49(11):2894–2897.

[Zhang and Abdulla, 2026] Zhang, G. and Abdulla, W. (2026). Transformers meet hyperspectral imaging: A comprehensive study of models, challenges and open problems. *Engineering Applications of Artificial Intelligence*, 169:113947.

[Zhang et al., 2022] Zhang, K., Li, Y., Zuo, W., Zhang, L., Van Gool, L., and Timofte, R. (2022). Plug-and-play image restoration with deep denoiser prior. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):6360–6376.

[Zhou et al., 2024] Zhou, P., Lv, Y., Zheng, Y., Zhou, J., and Chen, L. (2024). CTIS spectral image reconstruction technology based on slit-scanning architecture. *Optics Express*, 32(14):25276–25291.